

Introduction to SoC FPGAs

VLSI 2017

Microprocessors and Digital Systems Laboratory
National Technical University of Athens

Athens
May 2017

Presentation Outline

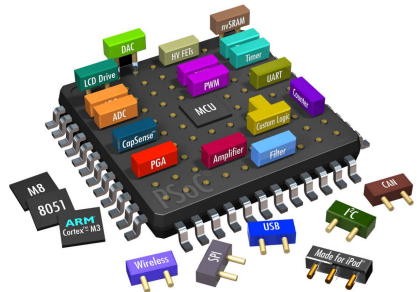
- 1 Introduction
- 2 The Zynq Family
- 3 Communication in SoC FPGAs
- 4 Demos

What is a SoC?

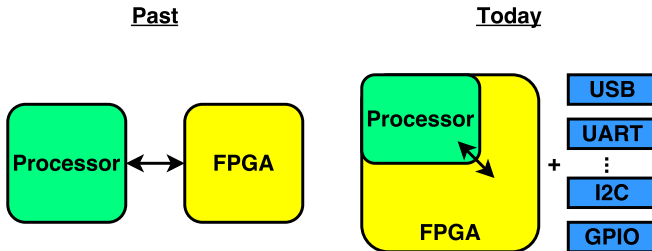
A complex IC that integrates digital, analog, radio frequency components (i.e. processor, hardware accelerators, ADCs/DACs, memory, interfaces and interconnection, peripherals etc.) all in a single chip

Benefits of a SoC device

- Reduced system cost
- Lower power consumption
- Smaller size
- Increased performance
- Better reliability



What is a SoC FPGA?



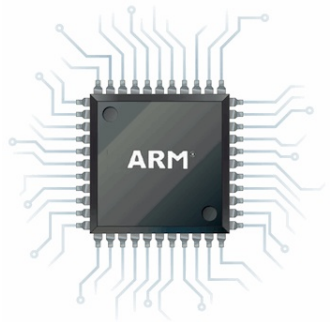
Benefits of a SoC FPGA device

- Lower communication overhead
- Improved size, power consumption
- Efficient HW/SW co-processing
- Increased productivity

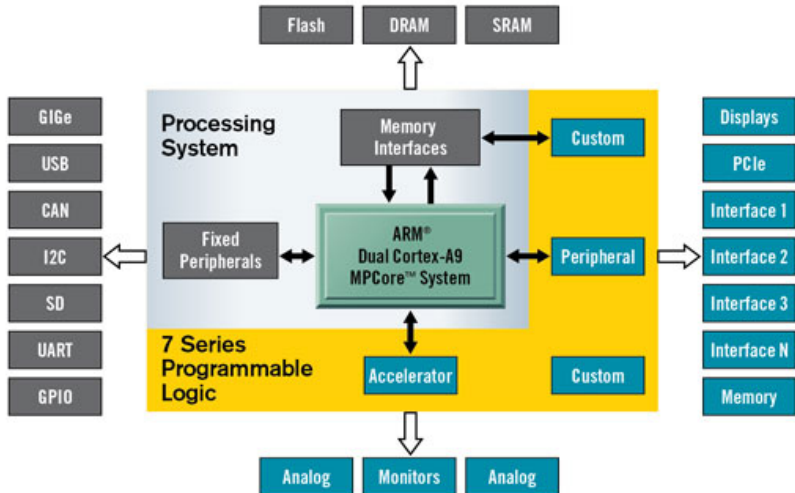
SoC FPGAs in the market

- **Xilinx**
 - Zynq 7000 AP SoC
 - Zynq UltraScale+ MPSoC
- **Intel** (bought Altera)
 - Stratix 10 SoC
 - Arria 10 SoC
 - Arria V SoC
 - Cyclone V SoC
- **Microsemi**
 - SmartFusion SoC
 - SmartFusion 2 SoC

All are ARM-based

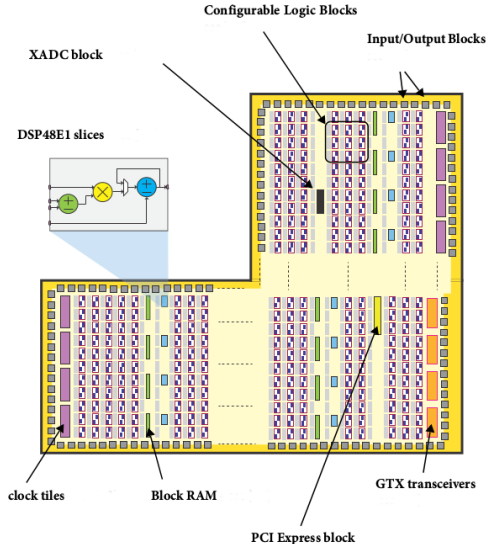


Zynq-7000 All Programmable SoC



Zynq Programmable Logic

- Configurable Logic Blocks (CLBs)
 - LUTs
 - DFFs
 - Multiplexers
 - Carry Chain Logic
- DSPs
- Block RAMs
- Input/Output Blocks
 - Gigabit Transceivers
 - XADC
 - PCI



The AMBA AXI Communication Protocol

For high-performance, high-frequency designs

- Master/Slave connection
- Address/control separate from data
- Separate write/read channels
- Unaligned transfers
- Single & burst based transactions
- Out-of-order transactions

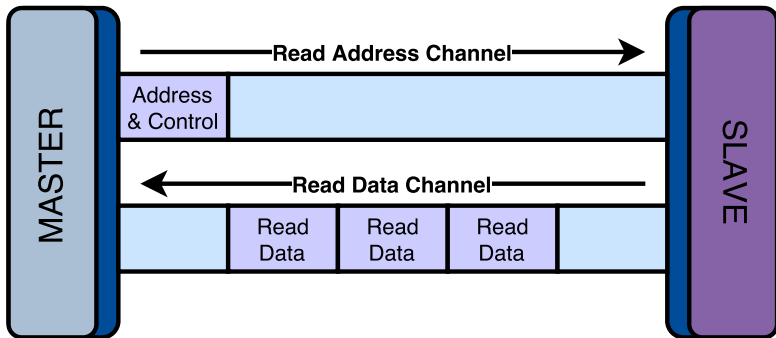
AXI4(-Lite)

- Memory mapped
- Bidirectional

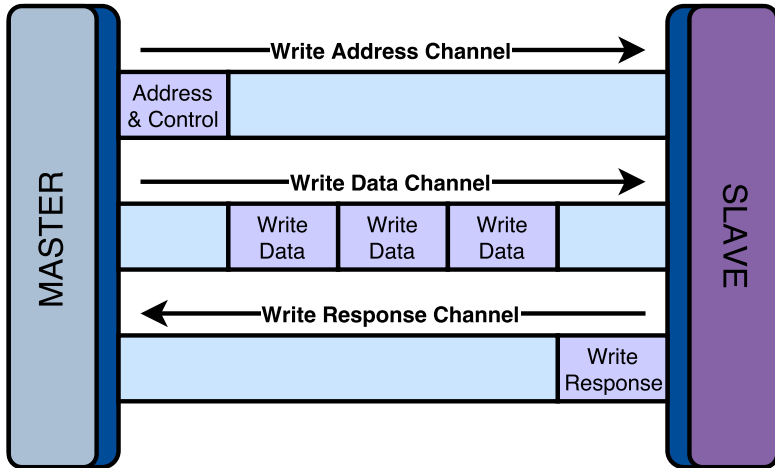
AXI4-Stream

- Non memory mapped
- Unidirectional

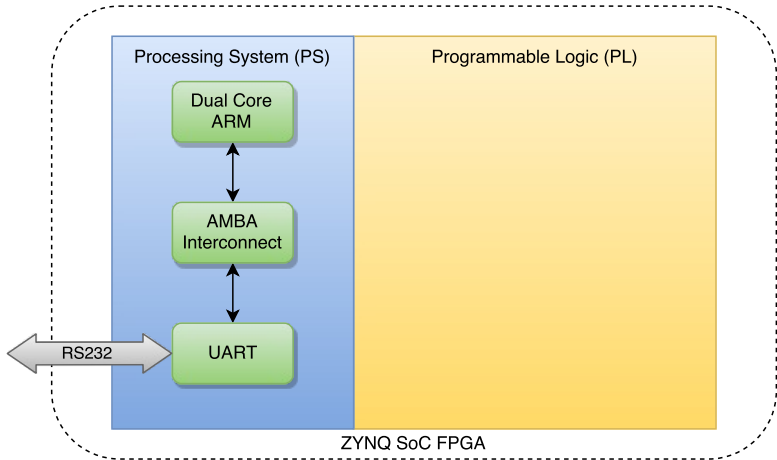
Read Channel Architecture



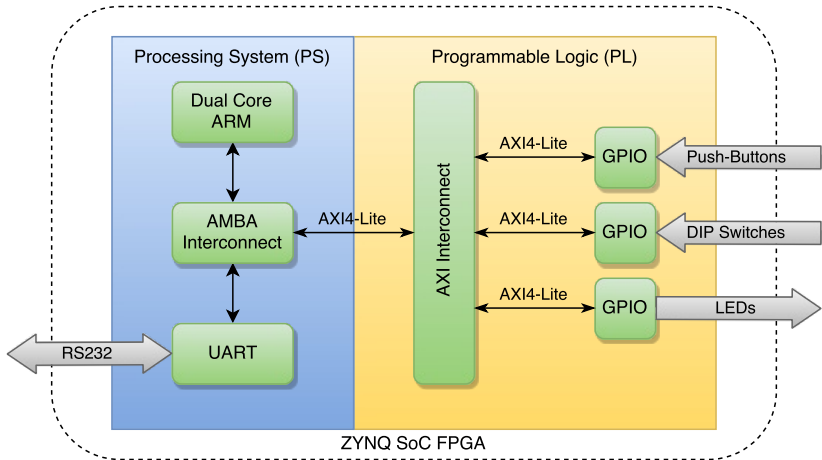
Write Channel Architecture



1st Demo : Using Zynq PS



2nd Demo : Communicating with External Devices



3rd Demo : Computing using the Zynq PL

